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SCHOOL OF COMPUTING AND INFORMATICS
DEPARTMENT OF INFORMATION TECHNOLOGY
BACHELOR OF SCIENCE IN INFORMATION SYSTEMS
MIS PROJECT IMPLEMENTATION
CIS 422

MASENO FOODS HUB
A WEB-BASED FOOD ORDERING SYSTEM FOR MASENO UNIVERSITY

**A Final Year Project Report Submitted In Partial Fulfilment of The Requirements For
The Award of The Degree of Bachelor of Science in Information Systems**

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YEAR: 2026

DECLARATION

I declare that this project titled **MASENO FOODS HUB**“*A Web-Based Food Ordering System for Maseno University Students*” is my original work and has not been presented for a degree or any other award in this or any other institution of higher learning. To the best of my knowledge, this project contains no material previously published or written by another person except where due reference has been made in the text.

Student Name: _____

Registration Number: _____

Signature: _____

Date: _____

Supervisor's Declaration

This project has been submitted for examination with my approval as the university supervisor.

Supervisor's Name: _____

Signature: _____

Date: _____

DEDICATION

I dedicate this project to my family for their continuous support, encouragement, and prayers throughout my academic journey. I also dedicate this work to my lecturers and friends whose guidance, motivation, and constructive criticism contributed significantly to the successful completion of this project. Finally, this project is dedicated to Maseno University students, whom this system aims to serve by improving access to efficient and convenient food ordering services within the campus.

ABSTRACT

The increasing student population at Maseno University has created significant pressure on campus food service providers, resulting in long queues, delayed service delivery, and inconvenience to students during peak hours. The traditional manual food ordering process requires students to physically visit food vendors, place orders, and wait for food preparation, leading to time wastage and disruption of academic schedules.

This project presents the design and development of **Maseno Foods Hub**, a web-based food ordering system developed to improve efficiency, convenience, and service delivery within Maseno University. The system enables students to browse available food menus, place orders online, and track order progress in real time. Food vendors can manage menus, receive and process orders, while system administrators oversee user management, vendor registration, and overall system operations.

The system was developed using **HTML, CSS, JavaScript, PHP, and MySQL**, providing a secure, reliable, and user-friendly platform accessible through standard web browsers. The implementation of the system minimizes congestion at food outlets, reduces waiting time, improves order accuracy, and enhances record management for vendors.

The study demonstrates that adopting digital food ordering solutions within university environments can significantly improve operational efficiency and user satisfaction. The system provides a practical technological solution tailored to the specific needs of Maseno University and offers a foundation for future enhancements such as mobile payment integration and mobile application support.

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LIST OF ABBREVIATIONS

Abbreviation	Full Meaning
ICT	Information and Communication Technology
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
PHP	Hypertext Preprocessor
SQL	Structured Query Language
DBMS	Database Management System
UI	User Interface
UX	User Experience
HTTP	Hypertext Transfer Protocol
API	Application programming interface
DFD	Data flow diagram
UML	Modelling language
URL	Uniform Resource Locator
JSON	JavaScript Object Notation
UAT	User Acceptance Terms

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CHAPTER ONE: INTRODUCTION

1.1 Background

The advancement of information and communication technology (ICT) has changed how different sectors deliver services to their users. Web-based systems have become essential tools for improving efficiency, accessibility, and user convenience. Digital platforms receive increasing adoption in higher learning institutions because they help educational institutions improve their operational processes while enhancing student learning experiences.

The student body at Maseno University constitutes a large and diverse group who depend on both on-campus cafeterias and adjacent food vendors to meet their daily dining needs. Currently, food ordering at the university is primarily conducted through manual, walk-in processes where students must physically visit food vendors to place orders and wait for food preparation. Students experience academic schedule disruptions and general campus life disturbances because this method leads to extended waiting times and site congestion during busy times.

Food vendors encounter business problems because they lack effective order management systems and their record keeping processes do not function properly and their sales tracking systems face difficulties and their demand forecasting capabilities remain restricted. The lack of a centralized digital platform prevents vendors from effectively delivering reliable service to students throughout their campus operations.

Students now need a web-based food ordering system at Maseno University because they have smartphones and internet connections which they use for their studies. The system gives students the ability to order food online which will lead to fewer people in cafeterias and better delivery of services. The project aims to create a web-based food ordering system that improves efficiency and convenience and customer satisfaction for both students and food vendors at Maseno University.

1.2 problem statement

Despite the rapid adoption of digital technologies in service delivery, food ordering at Maseno University remains largely manual. Students are required to physically visit food vendors to view menus, place orders, and wait for preparation, often resulting in long queues, delays, and missed academic commitments.

Food vendors also face operational challenges such as inefficient order handling, poor record management, and difficulty serving multiple customers during peak hours. The absence of a centralized digital ordering platform limits service efficiency, customer satisfaction, and proper coordination between students and vendors.

Therefore, there is a need for a web-based food ordering system that streamlines food ordering processes, reduces waiting time, improves operational efficiency, and enhances convenience for both students and food vendors at Maseno University.

1.3 objectives

1.3.1 General objective

To design and develop a web-based food ordering system that improves the efficiency and convenience of food ordering for Maseno University students.

1.3.2 Specific objective

- To design a user-friendly web interface for students to browse food menus and place orders online.
- To develop a system that enables food vendors to receive, manage, and process food orders efficiently.
- To implement an administrative module for managing users, vendors, and system operations.
- To develop a database for storing user information, food items, and order records.
- To evaluate the effectiveness of the system in reducing queues and improving service delivery.

1.4 Scope of the research

The study focuses on the design and implementation of a web-based food ordering system for Maseno University students. The system allows students to register, browse food menus, place orders, and track order status. Food vendors are able to manage food items, receive orders, and update order status, while administrators manage users and vendors. The system is accessible through web browsers and is limited to food vendors operating within Maseno University. Advanced features such as mobile applications and integrated online payment systems are beyond the scope of this study but may be considered for future enhancements.

1.5 limitations of the research

The study is limited to the development of a web-based system and does not include a mobile application. Internet connectivity issues may affect system accessibility for some users. Additionally, the system does not integrate online payment services such as mobile money due to time and resource constraints. User adoption and system effectiveness may also depend on the willingness of students and food vendors to embrace the technology.

1.6 Significance of the project

The study is justified by the need to improve efficiency and convenience in food service delivery at Maseno University. The proposed system will benefit students by saving time, reducing queues, and providing easy access to food ordering services. Food vendors will benefit from improved order management, accurate record keeping, and better customer service. The university administration may also benefit from improved coordination of campus food services.

Academically, this study contributes to the application of web-based technologies in solving real-world problems within higher learning institutions. It also provides a reference for future research and system development in campus-based service automation.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The chapter assesses current research on web-based food ordering systems which academicians have developed for educational institutions and commercial dining establishments. The review assesses existing food ordering systems through their benefits and challenges and their connection to related research. The study examines how systems boost operational efficiency and customer convenience and service delivery and what limitations those systems have when used in university environments.

The chapter describes all the works that have been studied while it reveals a research gap which exists because current systems have particular deficiencies. This gap establishes the need for a web-based food ordering system which will serve as the ordering system for food delivery for students at Maseno University. The review demonstrates how the study fits into existing research and shows how it will enhance campus food delivery systems through technological development.

2.2 Related work

The food service industry needs digital food ordering systems which existing studies and systems have developed to improve food delivery operations through better efficiency and easier customer access.

i. Online Food Ordering Platforms in Educational Institutions

Campus-specific food ordering systems enable students to order food online which leads to shorter waiting times and better service delivery. The university food ordering web application developed by Smith et al. (2020) enables students to access lunch menus and place orders while tracking their pickup times. The system improved student experience but lacked advanced order management features for vendors.

ii. General Food Delivery Systems

The commercial food delivery platforms used by Uber Eats Grubhub and Domino's online ordering system show high user adoption because they provide simple operation and

convenient service. The platforms offer real-time order tracking and multiple vendor selection and secure online payments according to Johnson and Lee 2019. The academic research demonstrates that food ordering systems boost customer satisfaction and operational efficiency when they function as part of vendor workflows.

iii. Mobile-Based Food Ordering Solutions

The mobile app-based food ordering systems which include Toast and Square for Restaurants deliver comprehensive interfaces that enable users to receive real-time order updates. The research conducted by Ahmed and Patel 2021 demonstrated that mobile platforms improve access to services however they create barriers for users who lack either smartphones or stable internet connections.

2.3 Related systems

This section discusses existing food ordering systems, highlighting their benefits and associated challenges.

2.3.1 Benefits of related systems

I. Convenience and Accessibility:

Web-based and mobile food ordering platforms allow users to place orders remotely at any time, saving time and reducing queues.

II. Efficiency:

Automated order processing improves coordination between customers and vendors, minimizes human errors, and streamlines service delivery.

III. Real-Time Updates:

Many systems provide order tracking features that enhance transparency, giving users status updates from preparation to delivery or collection.

IV. Improved Record Keeping:

Digital systems maintain order histories and sales records, which help vendors with analytics, forecasting, and planning.

V. Vendor Reach and Marketing:

Listing multiple vendors on one platform increases customer engagement and vendor visibility.

2.3.2 Challenges of Related Systems

I. Dependence on Internet Connectivity:

Users without reliable internet access may find it difficult to use web-based systems consistently.

II. Technical Complexity:

Some platforms require advanced backend infrastructure and maintenance, which may be costly for small vendors or campus environments.

III. User Adoption:

Resistance to technology among certain user groups may limit the effectiveness of the system.

IV. Payment Integration Issues:

Lack of reliable online payment options in some regions or institutions can affect system usability and vendor revenue collection.

VI. Security Concerns:

Ensuring data protection, secure authentication, and safe transaction processing remains a challenge in web-based applications.

2.4 Summary of Related Works

A review of literature shows that most existing food ordering systems focus on enhancing user convenience, reducing waiting time, and improving operational efficiency. Systems developed for educational institutions and commercial platforms share common objectives: simplifying the ordering process, providing real-time order updates, and maintaining digital records. While several studies demonstrate positive impacts, many systems either lack vendor management features or advanced functionalities such as integrated payments and analytics. Furthermore, research on food ordering specifically tailored for campus environments in Kenyan universities remains limited.

2.5 The general gap

The literature review together with system assessment shows three main gaps which need to be addressed.

A. Limited Campus-Focused Solutions:

The existing systems which primarily serve commercial food delivery services fail to meet the specific requirements of university students who need quick ordering options between their classes.

B. Incomplete Order Management for Vendors:

Some systems focus on enhancing customer experience yet they fail to deliver complete vendor dashboards which enable order processing and tracking.

C. Lack of Integrated Analytics and Reporting:

Only a few systems offer comprehensive sales reporting and analytics features which assist vendors and administrators in making data-driven decisions.

D. Limited Adoption of Local Payment Methods:

The user convenience and solution adoption rates suffer because many systems fail to support popular mobile payment methods which include M-Pesa in Kenya.

2.6 Conclusion

The reviewed literature confirms that web-based food ordering systems provide significant benefits such as improved service delivery, convenience, and efficiency. However, existing solutions often lack full functionality required in a campus context, including seamless vendor coordination, analytics, and integration with local payment services. Furthermore, limited research exists on systems tailored to universities in Kenya. This study proposes to address these gaps by designing and developing a web-based food ordering system specifically for Maseno University students, integrating student usability, vendor management, and administrative oversight within a single efficient platform.

CHAPTER 3: PROJECT METHODOLOGY

3.1 Introduction

This chapter describes the methodology used in the development of the web-based food ordering system for Maseno University students. It outlines the requirement gathering and analysis techniques, design tools and processes, and justifies the choice of tools used to ensure the system meets user needs and project objectives.

3.2 Requirements (Gathering and analysis)

Requirement determination involves identifying user needs, system expectations, and operational constraints to ensure that the proposed system effectively addresses the problem. In this study, several requirement gathering tools were used to collect accurate and reliable information from stakeholders.

3.2.1 Identification and Justification of Tools Used in Requirement Determination

The following tools and techniques were used to gather system requirements:

1. Interviews

Description:

Researchers conducted interviews through structured face-to-face dialogues with chosen participants from Maseno University as well as food vendors. Researchers developed a set of questions to facilitate discussions about existing problems in food ordering systems and the expected features of upcoming online platforms.

Purpose:

The interviews served the objective of gathering complete user experience information together with their difficulties and precise system needs which other research methods cannot capture.

Justification:

Interviews were selected because they provide direct user contact which helps researchers understand user needs while clarifying their responses to interviews. The researcher used this tool to collect both qualitative data and real-world insights which were obtained from food vendors and campus cafeteria users.

2. Questionnaires

Description:

The Maseno University questionnaire system collected standardized responses through its structured questions which were distributed to a larger student body. The researchers designed questions to assess how students currently order food and

which system features they prefer and how likely they would be to use an online ordering platform.

Purpose:

The purpose of questionnaires was to collect quantitative data from a wide range of users to identify common trends, preferences, and system expectations.

Justification:

The researchers used questionnaires because they enable fast and economical data collection from many people. The system requirements required statistical evidence which the researchers obtained from interview information validation.

3. Observation

Description:

The project team monitored all food ordering activities at campus cafeterias during their peak lunch and dinner periods.

Purpose:

The observation study aimed to find out practical issues which users encounter through their operational activities that they do not share during interviews or questionnaires.

Justification:

The observation method enables researchers to gather direct evidence about how systems function in their actual environments. The method enabled the researcher to observe actual work activities while he confirmed user-reported issues.

4. Document Review

Description:

The document review process required researchers to analyze all university-based ICT policies and project documentation and existing food ordering system records.

Purpose:

The document review process enables researchers to acquire background information and system standards knowledge and best practice identification for their proposed system.

Justification:

The tool enables researchers to obtain secondary data which helps validate primary research findings and verifies that the proposed system meets both institutional and technical requirements.

5. Internet and Literature Review

Description:

Researchers examined related journals and articles and online sources about web-based food ordering systems.

Purpose:

The research study aimed to analyze the current state of available technologies and operational difficulties which affect similar systems.

Justification:

The research team selected this method to prevent duplication of existing solutions while ensuring the system would use established design methods and tested technological solutions.

Conclusion

The combination of interviews, questionnaires, observation, document review, and literature review ensured comprehensive and accurate requirement determination. Using multiple tools allowed cross-validation of data, improved reliability, and ensured that the system requirements reflect the real needs of Maseno University students and food vendors.

3.2.2 Approach to analysis of requirement

The analysis of requirements was conducted using a systematic approach to ensure that the collected data was accurate, well-organized, and reflective of user needs. The process began with data collection and organization which continued with quantitative analysis of survey data and ended with multiple rounds of requirement improvements.

1. Data Collection and Organization

The research team collected data through interviews, questionnaires, observation, and document analysis. The researchers created organizational structures for their collected data which enabled them to conduct their analytical work more efficiently. The research team created three areas to classify questionnaire results which included common problems with ordering and preferred system features and expectations about system usability. The team transcribed interview and observation data to create summaries which showed the most important user problems and their related issues.

The research team designed their data organization system to create an efficient process for discovering patterns while they examined all necessary data during their requirement analysis work.

2. Quantitative Analysis for Survey Data

The research team used quantitative analysis methods to evaluate the data which they had gathered through questionnaires. The research team converted responses into coded data which they analyzed through basic statistical methods that included frequency counts and percentage calculations. The analysis identified the most frequent obstacles that students encountered along with the system features that students wanted most.

The researchers used tables and graphs to present their quantitative research findings which helped them make decisions. The system requirements were established by designating features which received higher response rates as mandatory requirements while features with lower response rates were classified as optional or future system enhancements.

3. Iterative Refinement of Requirements

The requirement analysis process conducted through its iterative method ensured that results would maintain both their correctness and their ongoing relevance. The team reviewed initial requirements which had emerged from data analysis. They refined these requirements through ongoing student and food vendor input. The team used feedback to delete requirements which lacked clarity or had contradictions.

The iterative refinement process enabled the team to create final requirements which matched user expectations and project goals while remaining practical and achievable. The system specifications received ongoing improvements throughout the development process until it reached its design and implementation phases.

Conclusion

The requirement analysis process used in this study established a methodical system for collecting requirements and analyzing them with precise measurements and ongoing requirement updates. The research study established an effective foundation for designing and developing the web-based food ordering system at Maseno University through its three combined methods of organized data collection and quantitative analysis and iterative refinement.

3.3 Design tools and processes

The design phase translates analyzed system requirements into a structured and implementable solution. This section describes the design tools, programming languages, and processes used in designing the web-based food ordering system, and justifies their selection.

3.3.1 Design tools and justification

a) HTML (HyperText Markup Language)

Description:

HTML provides the framework for web page design which enables users to create forms and menus and buttons as system elements.

Justification:

The web content structure of HTML serves as the standard language which provides web compatibility for all browsers.

b) CSS (Cascading Style Sheets)

Description:

CSS controls the style of system interface elements which includes designing the layout and choosing color schemes and creating responsive designs.

Justification:

The interface design of CSS enables users to navigate the system easily while delivering a responsive experience which enhances usability across various devices.

c) JavaScript**Description:**

JavaScript enables users to interact with the system while the system updates content dynamically through validation and cart operations and data loading without refreshing the page.

Justification:

JavaScript allows users to interact with the system through multiple activities which create better system performance while system responsiveness improves.

d) PHP (Hypertext Preprocessor)**Description:**

The server-side scripting language PHP controls system operations which include user authentication and order processing and database system communication.

Justification:

PHP was selected because it provides simple usage combined with MySQL database system which supports web applications through its extensive community backing.

e) MySQL**Description:**

MySQL functions as a relational database management system which stores user information and food menus and order details and transaction histories.

Justification:

MySQL delivers reliable data storage together with quick query operations and PHP system integration which makes it perfect for database requirements.

3.3.2. Tools Used in the Design of the Solution

The following tools were used to design and model the system before implementation:

a) Data Flow Diagrams (DFDs)

Description:

DFDs show the system data movement through user interactions with processes and data storage facilities.

Justification:

The DFDs serve as a visual tool which shows system processes and data connections to enhance both system understanding and implementation.

b) Entity Relationship Diagrams (ERDs)

Description:

The system defines database entities through ERDs which show their attributes and all connection types to users and vendors and food items and orders.

Justification:

ERDs confirm correct database design which maintains data standards through effective data processing methods.

c) Use Case Diagrams

Description:

Use case diagrams show system operations from the user perspective through user-system interaction representation.

Justification:

The system functions can be identified through these tools which also ensure that all user requirements receive proper attention.

d) Wireframes and Mockups

Description:

The system interfaces were built through wireframes and mockups which created their layout and navigation system.

Justification:

The tools create visual representations of user interfaces which help to build designs from early concepts while increasing usability and decreasing redesign expenses.

3. Design Process

The system design process used standard procedures which the following descriptions explain.

The process created system requirements which turned into documented system specifications.

The design established system architectural components which defined the organization of frontend and backend and database systems.

The system processes were modeled through DFDs and Use Case Diagrams.

The database design process used ERDs for database structure design.

The user interface design process used wireframes and mockups for system interface layout design.

The process made sure that the completed system design matched what users wanted and what the project needed to achieve.

4. Conclusion

The chosen design tools and programming languages and processes were selected to deliver a system which maintains operational efficiency and user satisfaction and system maintenance capabilities. The design and development process of the web-based food ordering system for Maseno University students received effective web technologies and visual modeling tools because they established a strong basis for standardized web technologies which enabled designers and developers to create the system.

CHAPTER FOUR: SOLUTION ANALYSIS AND SPECIFICATION

4.1 Requirement determination

Requirement determination involved identifying and documenting all relevant system requirements needed for the successful development of the web-based food ordering system for Maseno University students. This process was guided by information collected through interviews, questionnaires, observation, and document review.

4.1.1 Requirement determination process

The process to determine requirements used a systematic approach to locate and gather user needs and expectations which needed to be documented for the web-based food ordering system because it needed to solve problems at Maseno University. The process used established stages which achieved both complete and accurate results.

1. Identification of Stakeholders

The first step involved identifying key stakeholders who interact with or are affected by the system. The three groups of stakeholders consisted of Maseno University students and campus food vendors and system administrators. The process of identifying stakeholders enabled the collection of requirements from all groups of relevant users.

2. Data Collection

The data collection process employed various techniques for requirement gathering which included interviews and questionnaires and observation and document review. The study used interviews and questionnaires to gather user needs and expectations while observation showed how users currently order food. Document review helped establish system requirements which matched institutional policies and best practices.

3. Requirement Elicitation

Requirement elicitation involved the process of obtaining essential system requirements through the analysis of collected data. User responses were analyzed to identify common challenges, expectations, and desired system features. The data was transformed into system requirements which could be measured and evaluated.

4. Requirement Classification

The elicitation process resulted in two types of requirements which were functional requirements and non-functional requirements. The functional requirements explained how the system would perform its tasks which included food ordering and vendor management while the non-functional requirements specified the system's essential quality features which included usability and security and performance.

5. Requirement Documentation

The team documented all identified requirements through a structured documentation process. The documentation content functioned as a system design and implementation reference point which maintained design consistency while enabling requirement traceability.

6. Requirement Validation

The documented requirements were reviewed and validated with selected stakeholders to confirm accuracy and relevance. The user feedback process helped refine requirements because it clarified all ambiguous points and resolved conflicts between requirements.

7. Requirement Approval

The team proceeded to the final step which required stakeholders to approve all documented requirements. The approved requirements established the framework which guided system design and modeling and implementation.

Conclusion

The requirement determination process allowed the organization to identify all system requirements through a structured method which included validation and documentation. The requirement development process achieved accurate results because it engaged the stakeholders while using various data collection methods to identify critical system requirements needed for successful development of the web-based food ordering system for Maseno University students.

4.2 Relevant requirements identified

The project team found the necessary requirements after they studied the information which stakeholders provided, namely Maseno University students, food vendors, and system administrators.

The identification process began with the examination of data which researchers obtained through interviews, questionnaires, observation, and document review. The team investigated the manual food ordering system to create explanations of user needs and system requirements. The project team only accepted requirements which directly enhanced efficiency and convenience while improving service delivery.

The team established two categories of requirements which included functional and non-functional requirements. The functional requirements defined essential system functions which included user registration, food menu display, order placement, order tracking, vendor order management, and administrative control. The requirements became essential because they helped achieve the main purpose of creating an automated food ordering system for university users.

The non-functional requirements defined system quality attributes which included usability, security, reliability, and performance. The requirements became essential because they guarantee that students and food vendors can use the system daily without any security or functionality problems.

The selected stakeholders validated the identified requirements to confirm their accuracy and relevance. The team used feedback to develop the requirements which included removing any unnecessary or unrealizable features. The process verified that all requirements were practical and they matched project objectives while being appropriate for the Maseno University setting.

4.2.1: Functional requirements

These define what the system should do.

- The system shall allow students to register and log in.
- The system shall allow students to browse food menus from registered vendors.
- The system shall allow students to place food orders online.
- The system shall allow students to track the status of their orders.
- The system shall allow vendors to add, update, and remove food items.

- The system shall allow vendors to receive and process orders.
- The system shall allow administrators to manage users and vendors.
- The system shall generate basic reports for vendors and administrators.

4.2.2: Non-functional requirement

These define how the system should perform.

- The system shall be easy to use and user-friendly.
- The system shall be accessible through standard web browsers.
- The system shall ensure data security through authentication.
- The system shall be reliable and available during operational hours.
- The system shall provide acceptable response time for user requests.

4.2.3: operational requirement

Operational requirements define the conditions and constraints under which the web-based food ordering system will operate once it is deployed. These requirements ensure that the system functions effectively within the Maseno University environment and supports day-to-day operations for students, food vendors, and administrators.

1. System Accessibility

The system shall be accessible through standard web browsers such as Google Chrome, Mozilla Firefox, and Microsoft Edge. Users shall access the system through internet-enabled desktop computers and laptop devices and mobile devices.

2. User Roles and Responsibilities

The system shall support multiple user roles which include students and food vendors and administrators. The system enables users to access specific functions based on their assigned user role through permission and access rights.

3. Availability and Operating Hours

The system shall be available for use during campus food service operating hours. The scheduled maintenance times will be announced to users before the maintenance starts to reduce user interruptions.

4. Data Management and Storage

The system stores and manages user data together with food item data and order data and transaction data in a centralized database. The organization implements regular data backups as a measure to safeguard against data loss.

5. Security and Access Control

The system requires users to authenticate their identity through secure login credentials. The system uses access control mechanisms to protect user data from unauthorized access.

6. System Performance

The system processes user requests within acceptable response times which enables smooth operation during peak usage periods that occur at lunch and supper times.

7. Error Handling and Recovery

The system displays error messages to users when the system encounters failures or they enter incorrect information. The system uses recovery mechanisms to restore normal operations after it experiences disruptions.

8. Maintenance and Support

The system enables maintenance and update processes and improvement implementation to occur without causing significant operational interruptions. The system administrators handle performance monitoring and they also manage the resolution of all technical problems.

4.3 Requirement analysis

The team executed requirement analysis to develop system specifications from user requirements after they completed requirement identification for the online food ordering system designed for Maseno University students.

The analysis process began with a detailed review of the requirements gathered from students, food vendors, and administrators. The team examined each requirement to remove all uncertainties while deleting duplicate requirements and resolving existing conflicts. The process ensured that only correct and applicable requirements were included in the system design process.

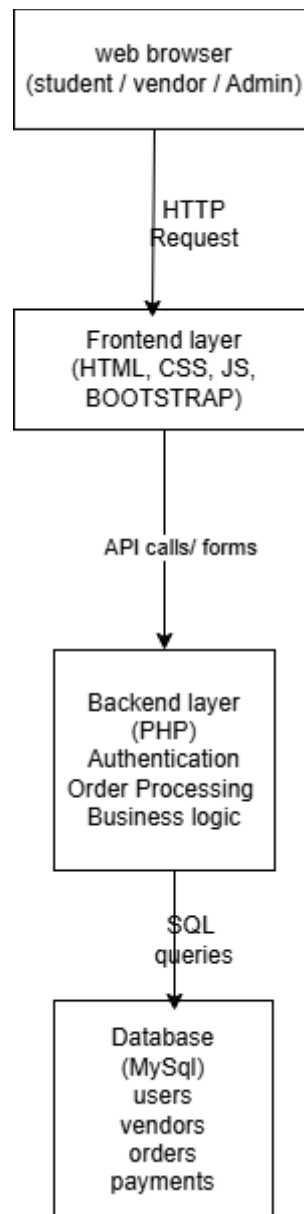
The team divided the examined requirements into functional requirements and non-functional requirements and operational requirements. The system core functionalities which included user registration and food menu display and order placement and order management were outlined in the functional requirements. The operational requirements described system usage parameters for the Maseno University environment while non-functional requirements specified system quality attributes which included usability and security and performance and reliability.

The evaluation process for requirements determination required the assessment of three factors which included technical feasibility and operational feasibility and economic feasibility. The team prioritized requirements which mapping to existing technologies could be executed within established time and resource constraints. The team documented less critical features for future system improvements.

The team represented the analyzed requirements through solution modeling techniques which included Data Flow Diagrams (DFDs) and Use Case Diagrams. The system models enabled users to visually understand both system processes and user interactions, which helped them confirm requirements and develop system design.

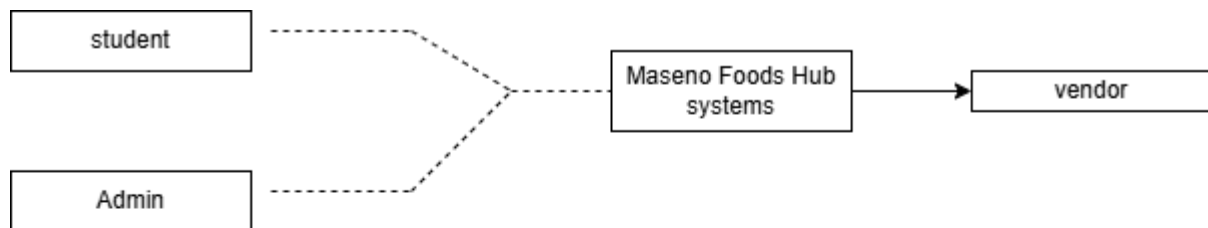
The team documented the refined requirements through clear writing, which they validated with selected stakeholders to confirm that the requirements reflected user needs and project objectives. The web-based food ordering system development process used structured requirement analysis to create design foundations for the system.

4.3.1: System architecture diagram



4.3.2: Data flow diagrams

Level -0 dfd

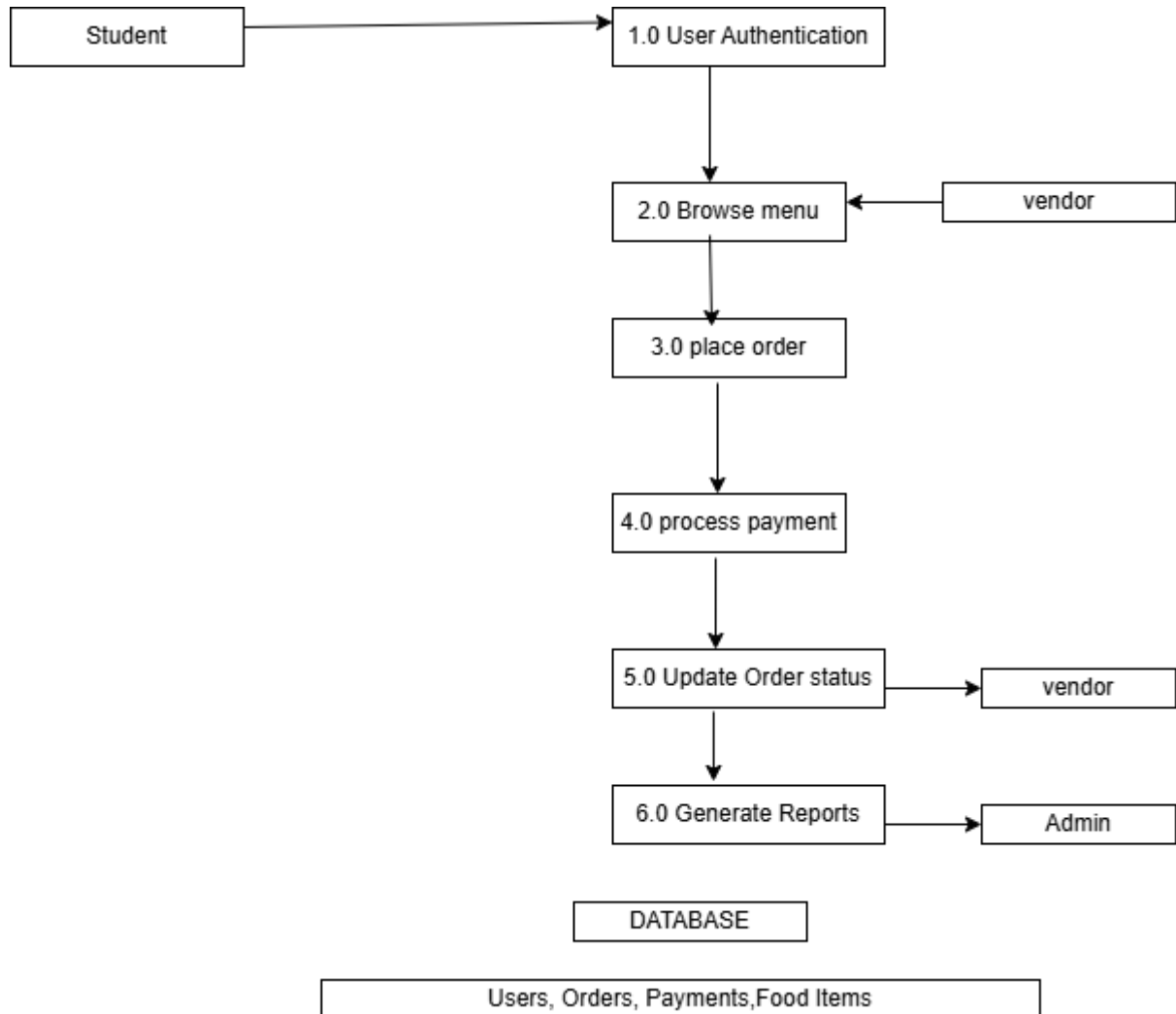


Data Flows:
Student → Order Request, Login Data
System → Order Status, Menu

Vendor → Menu Data, Order Updates
System → Order Notifications

Admin → User Management
System → Reports

Level 1 dfd



External Entities:

Student

Vendor

Admin

Main Processes

1.0 Manage Menu & Items

Handled mainly by vendors.

2.0 Place & Process Order

Core ordering flow (student → system → vendor).

3.0 Handle Payment

Processes transactions (often integrated with M-Pesa or card in Kenyan context).

4.0 Track & Update Order Status

Real-time updates (preparing, ready, picked up / delivered).

5.0 System Administration & Reporting

Admin-only oversight.

Data Stores (DS):

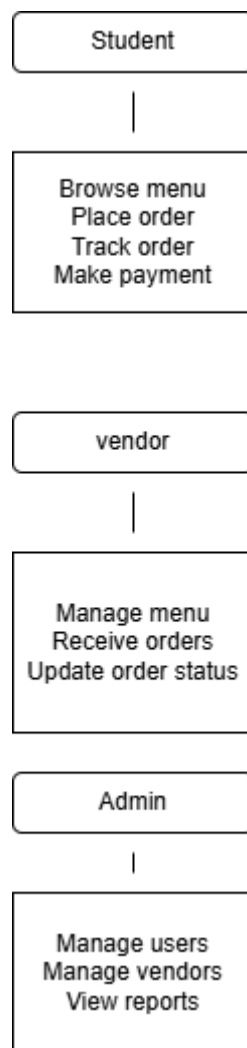
User Accounts (students, vendors, admin credentials & profiles)

Menu & Food Items (food list, prices, availability, vendor-linked)

Orders (order ID, items, student ID, vendor ID, status, timestamp)

Payments (transaction ID, amount, status, method)

4.3.2 Use case diagrams



This style shows:

Actors:

Student (Customer) – primary user ordering food.

Vendor (Restaurant/Cafeteria staff) – manages food and receives orders.

Sometimes a combined/delivery role or Admin.

Main Use Cases

View Restaurant Menu / Browse Menu

Search Restaurant / Search Food (by vendor or food type)

Place Order (student selects items, location e.g. hostel, notes)

Cancel Order

Track Order (view status: received → preparing → ready → picked up)

Receive Payment (handled by vendor or system)

Update Order Status (mostly Vendor: accept, preparing, ready, completed)

4.4 System requirements

Hardware and software requirements for *maseno foods hub* are described here under the categories of development, server deployment, and client (end user) use. These requirements ensure that an accessible, performant, and economical system in the rural context of Maseno university, where students mostly work with budget phones and variable connectivity, is met.

4.4.1 Hardware requirements

a) Development Phase

Computer with at least an Intel Core i5 processor or equivalent

Minimum of 8 GB RAM for smooth multitasking

At least 256 GB of storage space for source code and databases

Keyboard and mouse for system development

Reliable internet connection for research, updates, and testing

b) End-User Environment

Smartphone, laptop, tablet, or desktop computer

Minimum of 2 GB RAM

Screen capable of displaying web pages clearly

Internet connection (Wi-Fi or mobile data)

2. Software Requirements

a) Development Environment

Operating system such as Windows, Linux, or macOS

Code editor (e.g., Visual Studio Code)

Web server software (Apache via XAMPP)

Database management system (MySQL)

Programming languages:

HTML for structure

CSS for styling

JavaScript for interactivity

PHP for server-side processing

Web browsers for testing (Google Chrome, Mozilla Firefox)

Version control system (Git)

b) Client End-User Environment

Operating system such as Android, Windows, Linux, or iOS

Web browser (Google Chrome, Mozilla Firefox, Safari)

Internet connection to access the system.

CHAPTER FIVE: SOLUTION DESIGN

5.1: Introduction

This chapter presents the design of the **Maseno Foods Hub**, a web-based food ordering system developed for Maseno University. The design phase translates system requirements into a structured and implementable solution. It focuses on input design, data coding, input validation, database structure, system architecture, and output design.

The objective of this design is to ensure that the system is user-friendly, efficient, secure, and capable of handling real-time food ordering operations for students, vendors, and administrators.

5.2: Input Design

Input design refers to the process of creating forms and interfaces through which users enter data into the system. The aim is to ensure data is entered correctly, efficiently, and securely.

In the Maseno Food Hub system, several input forms were designed to capture information from different users including students, vendors, and administrators.

5.2.1: Main Input Forms

1. **Student Registration Form**

This form allows students to create accounts in the system. The fields include:

- Full Name
- Email Address
- Phone Number
- Password

2. **Vendor Registration Form**

This form allows vendors to register their food businesses in the system. The form captures:

- Vendor Name
- Business Name
- Email Address
- Phone Number
- Password

3. **Food Item Registration Form**

Vendors use this form to add food items to their menus. Fields include:

- Food Name
- Food Price
- Food Image

- Vendor ID

4. **Order Form**

Students place orders through this form. The system captures:

- Selected Food Item
- Vendor ID
- Pickup Location
- Payment Method
- Student Phone Number

These input forms were designed using **HTML, CSS, and Bootstrap** to ensure usability and responsiveness.

5.3 Data coding

Data coding is the assignment of unique identifiers to system entities to ensure efficient data management.

Data coding ensures that each entity in the system is uniquely identifiable. This improves data integrity, simplifies database operations, and enhances system performance.

The following coding scheme was used in the system:

ENTITY	CODING EXAMPLE
User ID	U001
Vendor ID	V001
Food Item ID	F001
Order ID	O001
Payment ID	P001

This coding approach ensures that each record in the system database can be uniquely identified

5.4: Input validation and input masks.

Input validation ensures that users enter correct and complete data before submission.

The Maseno Foods Hub system implements several validation techniques including:

1. **Required Field Validation**
Fields such as name, email, password, and phone number must be filled before form submission.
2. **Email Validation**
The system ensures that email addresses follow the correct format.
3. **Password Validation**
Passwords must meet minimum security requirements before account creation.
4. **Phone Number Validation**
Phone numbers must follow the Kenyan mobile phone number format.
5. **Numeric Validation**
Food prices must contain numeric values only.
6. **Server-side Validation (PHP)**
Prevents malicious input and ensures data integrity even if client-side validation is bypassed.

Validation was implemented using **JavaScript and PHP server-side validation**.

5.5: File / Database / Architecture Design

Main Database Tables

TABLE	DESCRIPTION
Users	Stores student, vendor, and admin accounts
Vendors	Stores vendor business information
Food_items	Stores food menu items
Orders	Stores students orders
Order_items	Stores ordered food items
Payments	Stores payment details

This structure ensures scalability, efficient data retrieval, and reduced redundancy.

5.6: System Architecture

The system follows a **three-tier architecture** consisting of:

1. **Presentation Layer**
Handles user interaction through web interfaces (HTML, CSS, JavaScript).
2. **Application Layer**
Implements business logic using PHP, including authentication, order processing, and validation.

3. **Data Layer**

Manages data storage using MySQL.

This architecture improves maintainability, scalability, and system security

5.7: **Output Design**

Output design focuses on how information is presented to users after processing.

The system generates different outputs depending on the user role.

5.7.1: Student Outputs

Students can view:

- Food menus
- Order confirmations
- Order status updates
- Payment information

5.7.2: Vendor Outputs

Vendors can view:

- Customer orders
- Student contact details
- Order status updates

5.7.3: Admin Outputs

Administrators can view:

- Vendor management reports
- System order reports
- User management information

All outputs are presented through a user-friendly web interface designed with Bootstrap

5.8: **Report Design**

The system generates several reports to support decision-making and monitoring.

5.8.1: Key Reports Generated

1. **Order Report**
Displays all orders placed by students including vendor name, food item, payment method, and order status.
2. **Vendor Report**
Lists registered vendors and their status (approved, suspended, pending).
3. **Payment Report**
Displays payment transactions made through the system.
4. **Food Items Report**
Shows all food items registered by vendors.

These reports help administrators monitor system usage and vendor activities.

5.8.2: Relevance of the Reports

Reports play a critical role in the Maseno Foods Hub system by transforming raw data into meaningful information that supports decision-making, monitoring, and system control.

1. Operational Monitoring

Reports enable administrators and vendors to monitor daily system activities such as orders placed, completed transactions, and system usage.

This helps ensure smooth operation and quick identification of issues.

2. Decision Making

Reports provide summarized data that supports informed decision-making.

For example:

- Vendors can identify best-selling food items
- Administrators can evaluate vendor performance

This allows stakeholders to make data-driven improvements.

3. Financial Tracking

Payment reports help track all transactions made within the system.

This ensures:

- a. Transparency
- b. Accountability
- c. Easy financial reconciliation

4. Performance Evaluation

Reports allow evaluation of system performance and user activity.

For instance:

- Number of orders per day
- Peak usage times

This helps in system optimization and future planning.

5. Accountability and Control

Reports ensure that all system activities are recorded and traceable. Administrators can track:

- Vendor activities
- User behavior

This improves system control and reduces misuse.

6. Data Analysis and Forecasting

Historical reports allow analysis of trends over time. This helps vendors:

- Predict demand
- Plan inventory
- Reduce food wastage

CHAPTER SIX: SOLUTION IMPLEMENTATION AND RECOMMENDATIONS

6.1: Solution/prototype Functionalities

6.1.1: Description of Functionalities

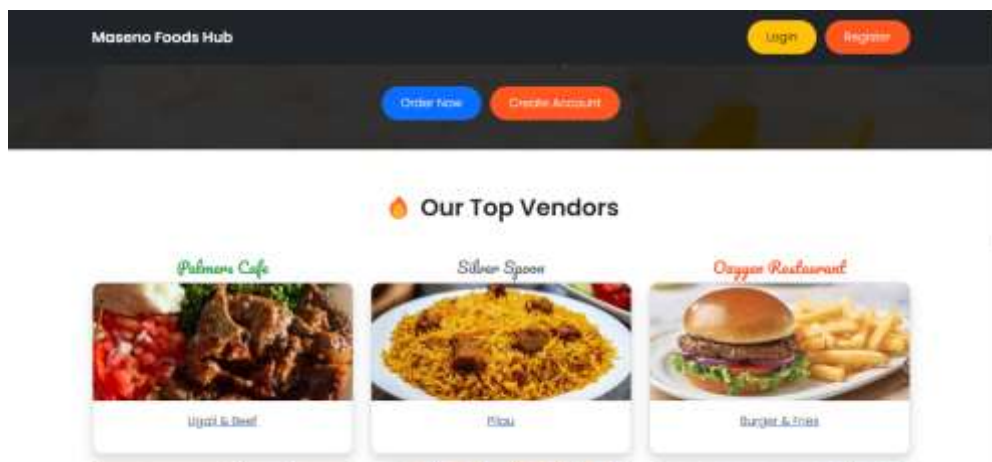
The **Maseno Foods Hub** system was successfully implemented as a fully functional web-based application designed to support food ordering operations within Maseno University.

The system consists of three main user modules:

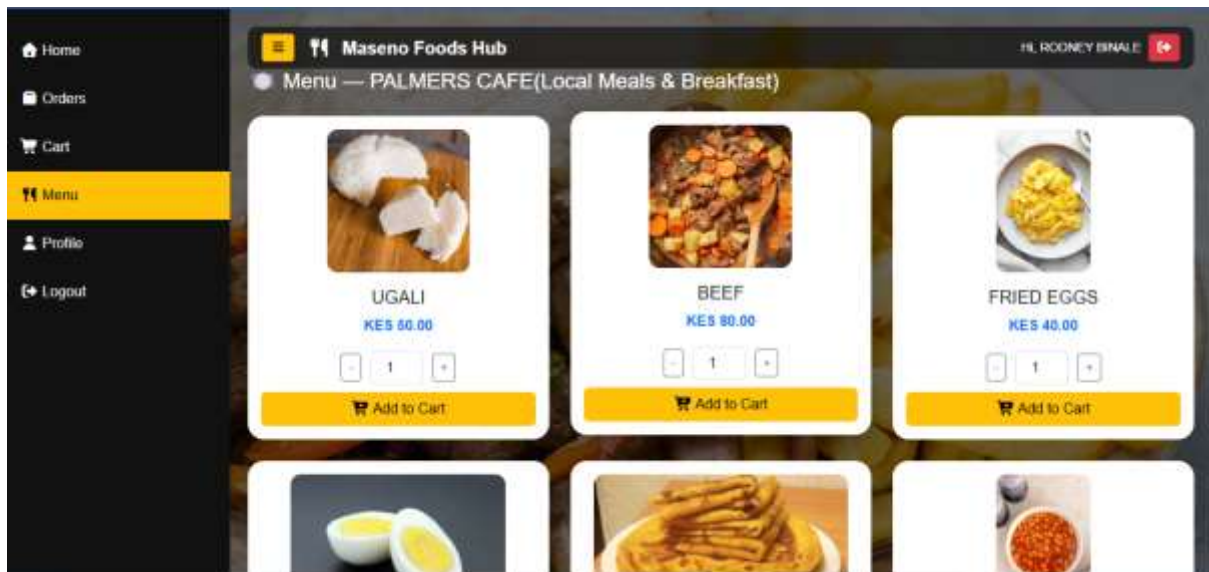
1. Student Module

The student module allows users to:

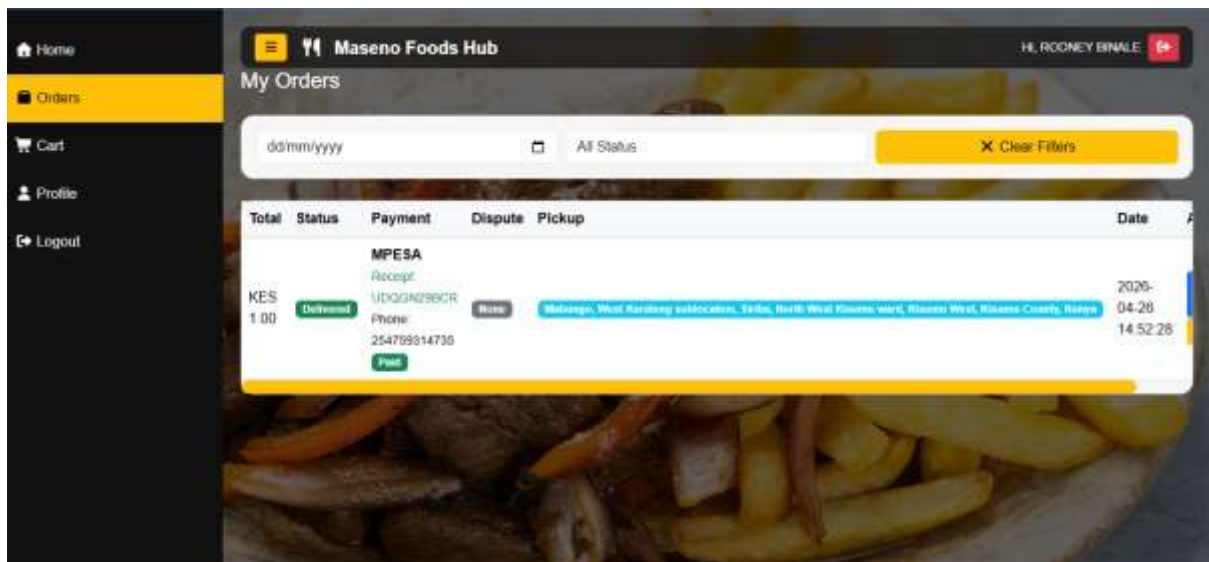
- Register and log into the system
- Browse available food menus from different vendors
- Place food orders
- Select payment methods (e.g., cash on pickup)
- Track order status in real time



Login page



Menu page at palmers café

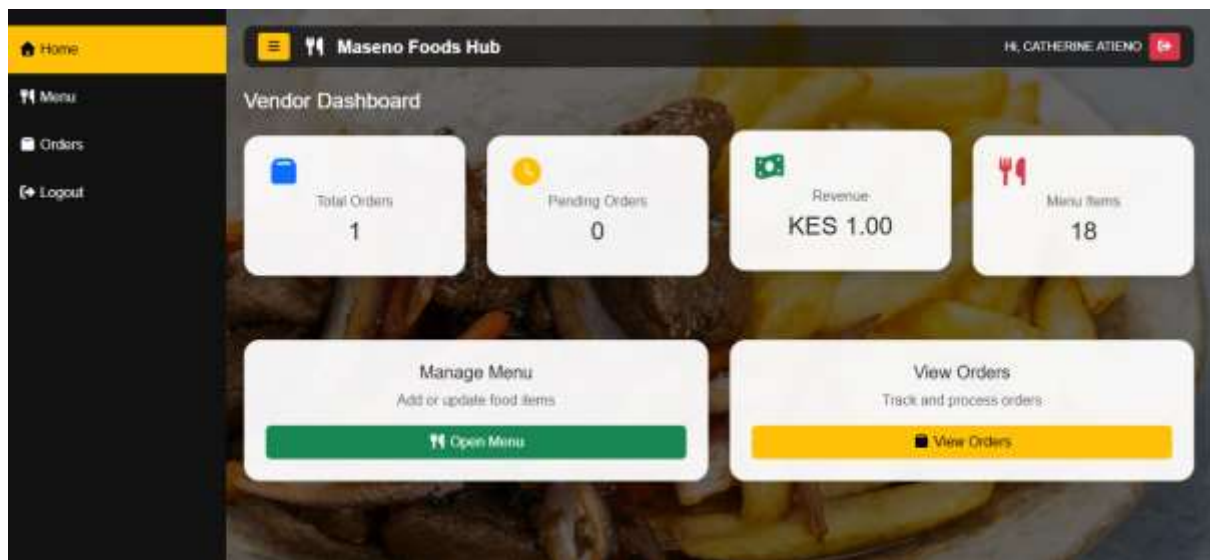


Orders page

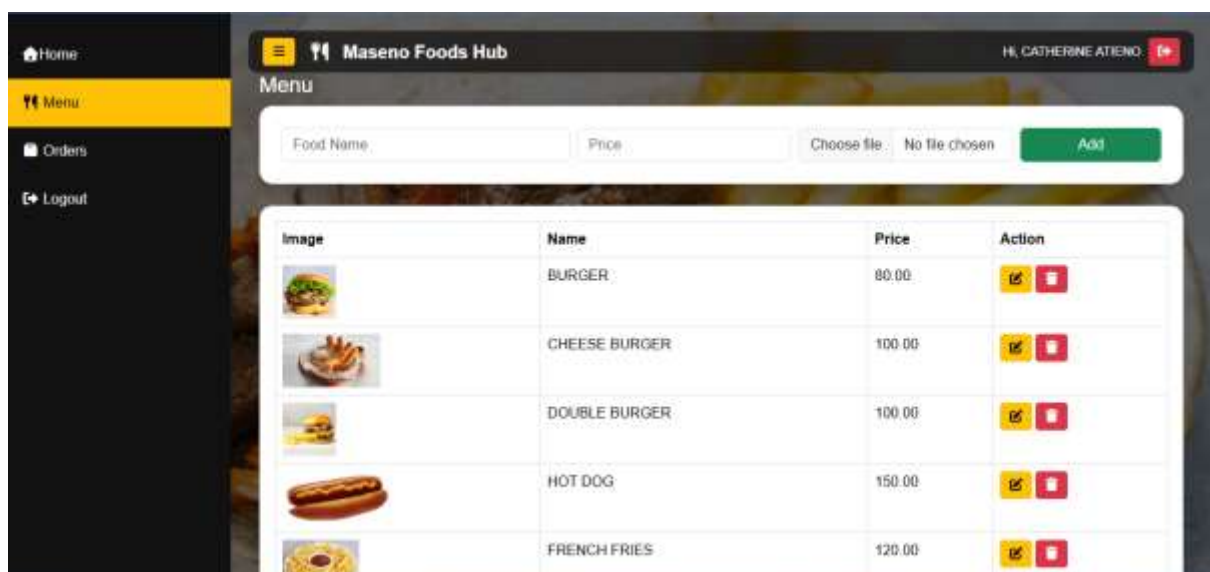
2. Vendor Module

The vendor module enables food vendors to:

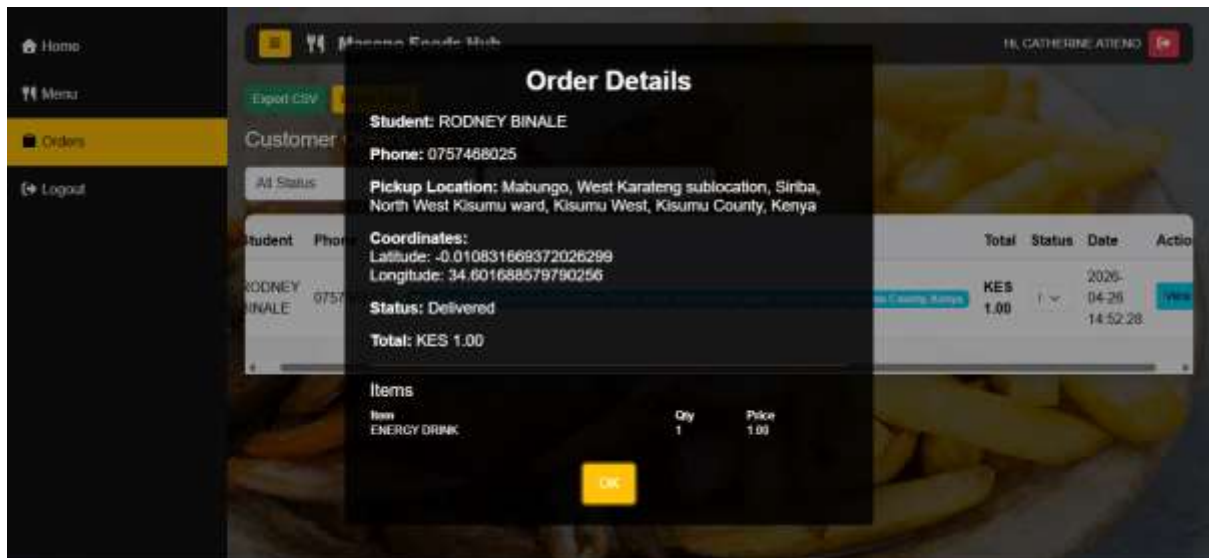
- Register and manage their profiles
- Add, update, and delete food items
- Receive and process incoming orders
- Update order status (e.g., preparing, ready, completed)



Vendor dashboard



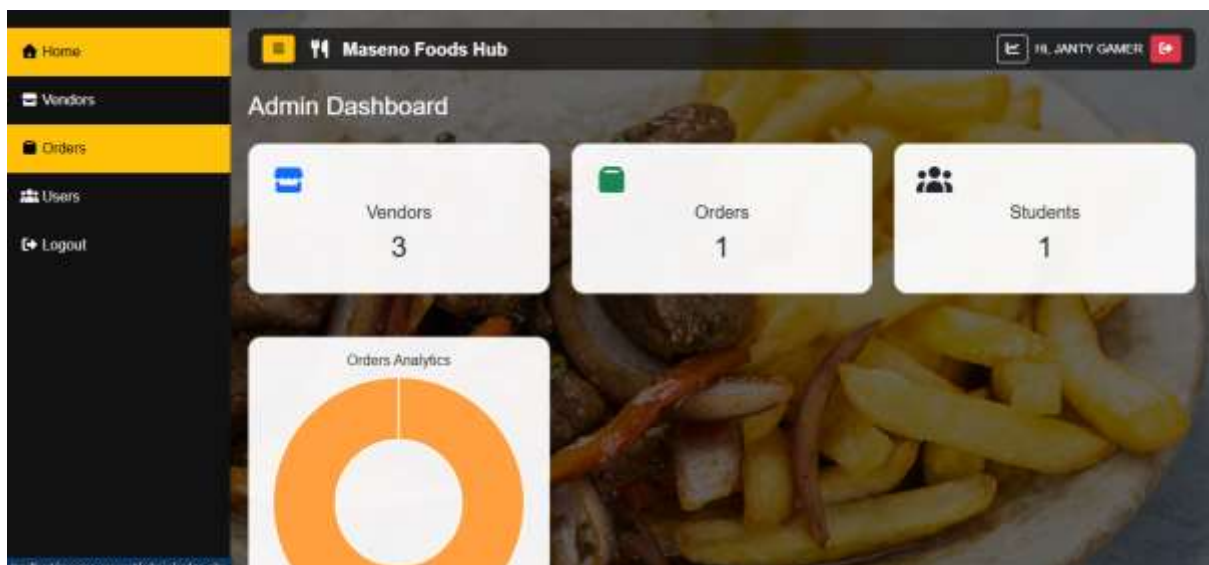
Add food



3. Administrator Module

The admin module provides system control functionalities such as:

- Managing users and vendors
- Monitoring system activities
- Viewing reports (orders, vendors, payments)



Admin dashboard



Reports page

6.1.2: Demonstration of Completeness

The prototype demonstrates completeness by successfully integrating all core functionalities required for a food ordering system.

These include:

- User authentication (login/registration)
- Menu management
- Order placement and processing
- Order tracking
- Basic reporting

The interaction between modules ensures that the system operates as a unified platform, meeting the objectives defined in earlier chapters.

6.2: Security Principles

6.2.1: Demonstration of Security Features

The system incorporates several security mechanisms to ensure data protection and secure system usage:

1. User Authentication

- Secure login using username and password
- Role-based access (student, vendor, admin)

2. Password Protection

- Passwords are stored securely (hashed)
- Prevents unauthorized access

3. Input Validation

- Both client-side (JavaScript) and server-side (PHP) validation implemented
- Prevents invalid or malicious input

4. Session Management

- Sessions used to maintain user login state
- Prevents unauthorized access to restricted pages

5. Database Security

- Structured queries used to prevent SQL injection
- Access to database restricted through backend logic

6.3: Testing and Associated Procedures

6.3.1: Testing Procedures and Levels

Testing was conducted to ensure that the system functions correctly, efficiently, and securely.

1. Unit Testing

Individual components were tested independently.

Examples:

- Login functionality
- Form validation
- Order submission

2. Integration Testing

Different modules were tested together to ensure proper interaction.

Example:

- **Student places order → Vendor receives order**

3. System Testing

The entire system was tested as a complete unit to verify that all functionalities work together correctly.

4. User Acceptance Testing (UAT)

End users (students and vendors) tested the system to confirm usability and functionality.

Sample Test Case Table

Test Case	Input	Expected Output	Actual Output	Status
Login	Valid credentials	User logged in	Successful	Pass
Login	Invalid password	Error message	Error shown	Pass
Place Order	Select item	Order created	Order saved	Pass
Add Food	Valid data	Item added	Success	Pass

6.4: Solution Documentation

6.4.1: User Help Manual

The system includes a user guide to assist users in navigating and using the system effectively.

a. Student Guide

- Register account
- Log in
- Browse menu
- Place order
- Track order

b. Vendor Guide

- Log in
- Add food items
- Manage orders

c. Admin Guide

- Manage users
- View reports

The documentation ensures ease of system adoption and reduces the learning curve for new users.

6.5: Presentation and Demonstration

6.5.1: Originality, Ownership and Relevance

The Maseno Foods Hub system is an original project developed to address a real-world problem within Maseno University.

- The system was designed and implemented independently
- It solves the problem of long queues and inefficient food ordering
- It is relevant to students, vendors, and university operations

The system demonstrates practical application of information systems concepts in solving real-life challenges.

6.6: Conclusion and Recommendations

6.6.1: Overall Quality of the Prototype

The developed system successfully meets its objectives by providing an efficient, user-friendly, and reliable food ordering platform.

Key achievements include:

- Reduced waiting time
- Improved order management
- Enhanced user convenience

6.6.2: Suggestions for Further Work

Although the system is functional, several improvements can be made:

1. Development of a mobile application
2. Implementation of delivery tracking
3. Advanced analytics and reporting
4. Notification system (SMS or email alerts)
5. Revenue Model and Commission System

6.7: Citations and References

Chauhan, V., & Gupta, R. (2021). Online food ordering systems and their impact on customer satisfaction. *International Journal of Information Systems*, 8(2), 45–60.

Lee, S., & Park, J. (2020). Improving operational efficiency through digital ordering platforms. *Journal of Business Technology*, 12(1), 23–35.

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Smith, A., Brown, D., & Clark, P. (2020). Web-based systems in educational institutions. *Journal of Information Technology Education*, 9(1), 34–50.